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标题	Computational, Mechanistic, and Experimental Insights into Regioselective Catalytic C – C Bond Activation in Linear 1-Aza-[3]triphenylene
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摘要（原文节选）

C – C bond activation by transition metal complexes in ring-strained compounds followed by annulation with unsaturated compounds is an efficient approach to generate structurally more complex compounds. However, the site of catalytic C – C bond activation is difficult to predict in unsymmetrically substituted polycyclic systems. Here, we report a study on the (regio)selective catalytic cleavage of selected C – C bonds in 1-aza-[3]triphenylene, followed by annulation with alkynes, forming products with extended π -conjugat ⋯

数据表 · Table 1（4 行，含数值行 4 行）

Relative Free Energy (kcal · mol – 1) for the Four Different Isomers Calculated at Two Different Temperatures, 298.15 and 443.0 K, Respectively

isomer	compound	relative free energy (kcal · mol – 1, 298.15 K)	relative free energy (kcal · mol – 1, 443.0 K)
1-proximal	2	+3.3	+3.6
1-distal	3	+0.9	+1.4
2-proximal	4	+0.09	+0.13
2-distal	5	0	0

正文对应句（逐字）：The starting 7,8-bis(trimethylsilyl)1-azatriphenylene 1a was prepared according to the procedure described by Vollhardt et al.,29 with small modifications from 5,6-bis(trimethylsilyl)-1-azabiphenylene, in an overall yield of 40%. Its nonsilylated analogue 1b was prepared by desilylation of 1a with C

数据表 · Table 2（14 行，含数值行 12 行）

Catalytic Insertion Reaction of Diphenylethyne and 4-Octyne with 1a

entry	catalyst	ligand	R1	(equiv)	yieldsa,bc			
4aa (%)	5aa (%)	C.Y. (%)b	6aa (%)	7aa (%)	C.Y. (%)b			
1d	[Ir(cod)Cl]2	dppe	Ph	2	<3	<3		0
2		dppe	Ph	1	8	24	32	0
3		dppe	Ph	2	0	0		25
4		dppe	Ph	10	0	0		22
5		dppp	Ph	2	8	12	20	4
6		dppp	Ph	10	10	10	20	8
7	[Rh(cod)Cl]2	dppe	Ph	2	0	0		17
8		dppe	Ph	10	0	0		26
9		dppp	Ph	2	<3	<3		<3
10e	[Rh(cod)2]BF4	dppef	Ph	10	15	21	36	<3
					4ab (%)	5ab (%)		6ab (%)
11	[Ir(cod)Cl]2	dppe	n-Pr	1	14	20	34	0
12		dppe	n-Pr	10	0	0		26

脚注（逐字）：aIsolated yields.bCombined yields.c<3% denotes the amount of the product that was detected by TLC but could not be isolated in sufficient purity.dOil bath.eTHF instead of toluene.f10 mol %.

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